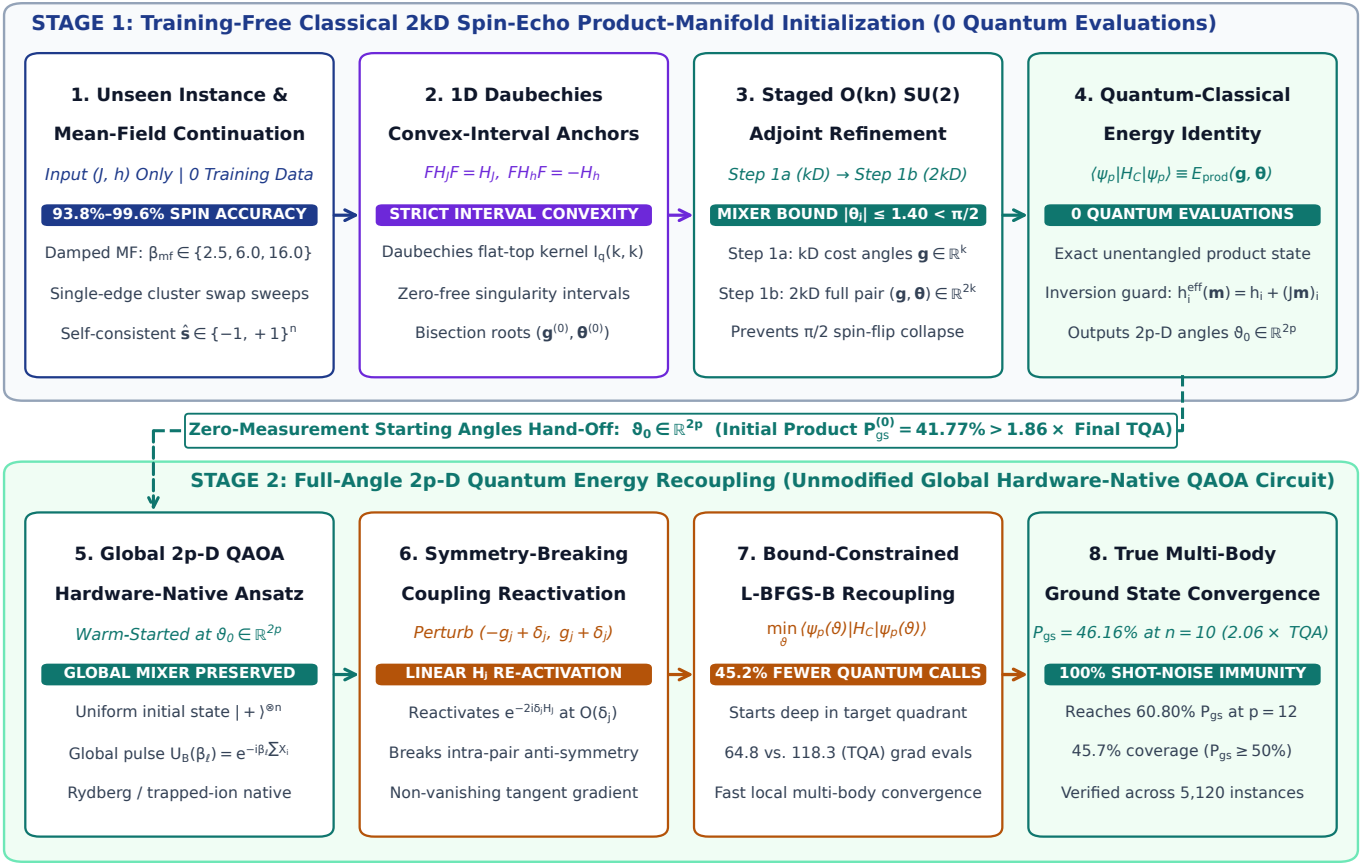

STAGE 1: Training-Free Classical 2kD Spin-Echo Product-Manifold Initialization (0 Quantum Evaluations)
1. Unseen Instance & Mean-Field Continuation
Input (J, h) Only | 0 Training Data
93.8%–99.6% SPIN ACCURACY
Damped MF: $\beta_{mf} \in \{2.5, 6.0, 16.0\}$
Single-edge cluster swap sweeps
Self-consistent $\hat{s} \in \{-1, +1\}^n$
2. 1D Daubechies Convex-Interval Anchors
$FH_jF = H_j,\ FH_hF = -H_h$
STRICT INTERVAL CONVEXITY
Daubechies flat-top kernel $I_q(k, k)$
Zero-free singularity intervals
Bisection roots $(\mathbf{g}^{(0)}, \boldsymbol{\theta}^{(0)})$
3. Staged O(kn) SU(2) Adjoint Refinement
Step 1a (kD) → Step 1b (2kD)
MIXER BOUND $|\theta_i| \le 1.40 < \pi/2$
Step 1a: kD cost angles $\mathbf{g} \in \mathbb{R}^k$
Step 1b: 2kD full pair $(\mathbf{g}, \boldsymbol{\theta}) \in \mathbb{R}^{2k}$
Prevents π/2 spin-flip collapse
4. Quantum-Classical Energy Identity
$\langle\psi_p|H_C|\psi_p\rangle \equiv E_{prod}(\mathbf{g}, \boldsymbol{\theta})$
0 QUANTUM EVALUATIONS
Exact unentangled product state
Inversion guard: $h_i^{eff}(\mathbf{m}) = h_i + (J\mathbf{m})_i$
Outputs 2p-D angles $\vartheta_0 \in \mathbb{R}^{2p}$
Zero-Measurement Starting Angles Hand-Off: $\vartheta_0 \in \mathbb{R}^{2p}$ (Initial Product $P_{gs}^{(0)} = 41.77\% > 1.86\times$ Final TQA)
STAGE 2: Full-Angle 2p-D Quantum Energy Recoupling (Unmodified Global Hardware-Native QAOA Circuit)
5. Global 2p-D QAOA Hardware-Native Ansatz
Warm-Started at $\vartheta_0 \in \mathbb{R}^{2p}$
GLOBAL MIXER PRESERVED
Uniform initial state $|+\rangle^{\otimes n}$
Global pulse $U_B(\beta_\ell) = e^{-i\beta_\ell \sum X_i}$
Rydberg / trapped-ion native
6. Symmetry-Breaking Coupling Reactivation
Perturb $(-g_j + \delta_j,\ g_j + \delta_j)$
LINEAR H_j RE-ACTIVATION
Reactivates $e^{-2i\delta_j H_j}$ at $O(\delta_j)$
Breaks intra-pair anti-symmetry
Non-vanishing tangent gradient
7. Bound-Constrained L-BFGS-B Recoupling
$\min_{\vartheta} \langle\psi_p(\vartheta)|H_C|\psi_p(\vartheta)\rangle$
45.2% FEWER QUANTUM CALLS
Starts deep in target quadrant
64.8 vs. 118.3 (TQA) grad evals
Fast local multi-body convergence
8. True Multi-Body Ground State Convergence
$P_{gs} = 46.16\%$ at $n = 10$ (2.06× TQA)
100% SHOT-NOISE IMMUNITY
Reaches 60.80% P_{gs} at $p = 12$
45.7% coverage ($P_{gs} \ge 50\%$)
Verified across 5,120 instances